Interactive Exhibits, Gallery & Evaluation

Team Member: Taki

Role: Interactive Exhibits, Presentation & Evaluation Lead

1. Contribution Overview

My part of the Neural Network Safari focused on making the project easy to follow and showing what the notebook actually does. Instead of building the CNN and LSTM, I focused on how students explore the exhibits, how the safari comes together, and how to explain the recorded results.

The main pieces were the animal–architecture activity, the fallback renderer that displays the Spider/Graph Neural Network (GNN) exhibit, and the gallery that brings all six animal–architecture pairings together. The classification results also helped connect the visual explanations to actual model performance.

My goal was to make the safari clear and engaging without making it seem like the notebook does more than it shows. This report separates saved outputs from features that still need another run or an interaction test.

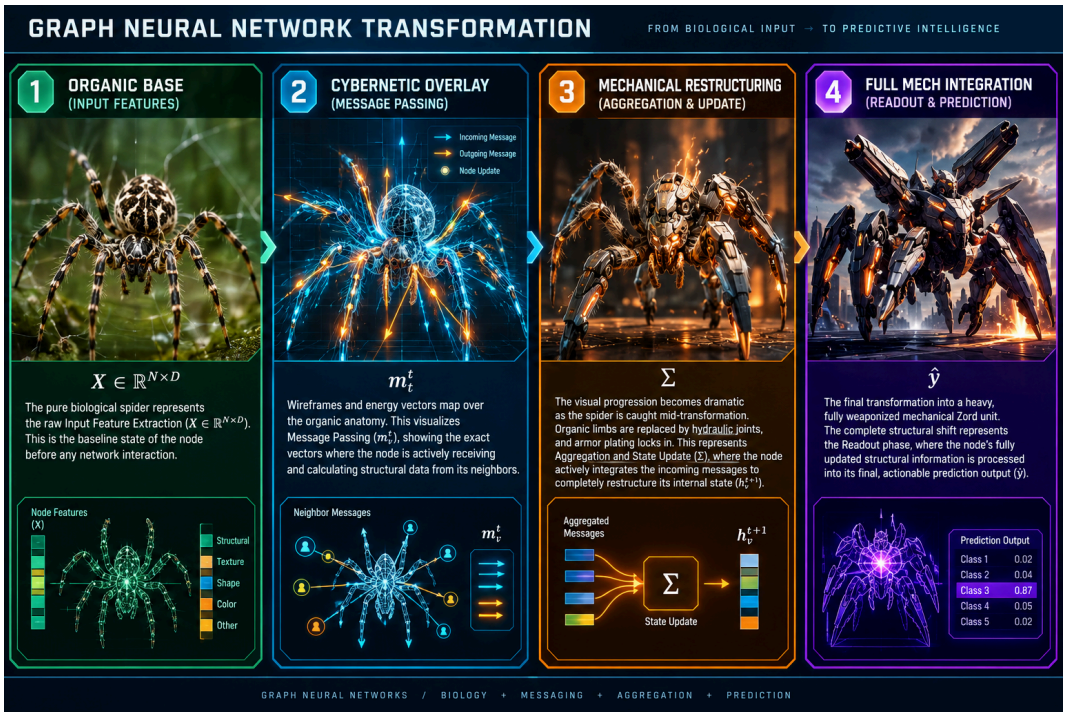
2. Interactive Exhibit Workflow

2.1 Organizing the Learning Activity

The activity offers two options: Spider/GNN and Honeybee/Transformer. Each includes four ordered steps and two extra statements associated with other network types. The idea is to help students recognize how information moves through a network, rather than simply match an animal to its name.

For the Spider, the sequence moves from input features to messages, then aggregation and a final readout. The animal theme gives students a familiar starting point, but the important part is understanding the process behind the analogy. The Honeybee is an additional activity option, not one of the six gallery images.

The sequence below illustrates a spider's transformation from an organic creature into a mech-like form across four stages.



This activity uses the animal metaphor to represent a computational process, rather than relying on appearance alone. Learners are expected to connect the analogy to an ordered mechanism. The Honeybee/Transformer is an additional activity option; it is not one of the six image assets explicitly listed in the gallery.

3. Presentation & Evaluation

The Python fallback displays the selected Spider exhibit, while the gallery places six animal–architecture pairings in a three-row, two-column layout. These outputs support the educational tour; they are not generated samples from a trained image model.

The recorded classification results provide a separate kind of evidence. They help connect the safari's explanations to model performance, but should not be treated as proof that every exhibit or interactive feature works.

4. Limitations, Reflection & Evidence

4.1 What Still Needs Checking

The image-generation code appears after the gallery code, even though the gallery needs those images. A saved gallery therefore does not guarantee that a fresh top-to-bottom run will reproduce the same output. Generating and checking the images first is a suggested improvement, not a repair I am claiming is complete.

The interaction also needs its own test. The saved Spider fallback shows that the Python-side renderer produced an output. It does not prove that a learner selected an option and completed the browser activity. That would require checking the selection, interface action, and resulting exhibit together.

4.2 What I Took Away

This part of the project showed me that looking good and working reliably are different things. The gallery makes the safari more engaging, but it still needs complete images, readable labels, and cells that run in the right order.

I also learned to be careful about how I describe the evidence. A saved output shows what appeared during an earlier run; it does not confirm that everything works in a fresh session. Keeping that difference clear makes the report more honest and gives the team specific things to test next.

4.3 Supporting Evidence

The relevant notebook sections are:

- **Cell 6:** Activity definitions, callback setup attempt, and saved Spider fallback output.
- **Cell 10:** Evaluation code, recorded accuracy results, and the training-loss plot.
- **Cell 11:** Gallery code and saved gallery image.
- **Cell 14:** Image-generation code and output messages.

These references use physical notebook order, starting at 1, rather than the execution numbers beside code cells.

4.4 Report Scope

This report uses the code and outputs already available in the notebook. It does not claim a new training run, a fresh-runtime test, or a verified browser interaction. Individual responsibility for the described work should be confirmed before submission; the notebook alone does not establish code authorship.